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Probabilistic Casing Collar Locator for Untethered Downhole Tools

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Abstract

The residual magnetic field amplitude shows significant variations along the length of the casing joints and usually a sharper variation occurs around the collars. These can be detected with magnetometers instead of conventional casing collar locators (CCL). However, magnetic signals within the casing joints can mask the signals around the collars. Here we test a probabilistic method that detects casing collars using hydrostatic pressure and the residual magnetic fields of casings. The method uses hydrostatic pressure measurement along with a Kalman filter to measure depth. If a magnetic field peak is observed, the algorithm compares the measured depth of the detected peak with the expected depth of a collar from the casing tally. If the two depths are relatively close, higher confidence is assigned to the detected collar location. Low confidence peaks are discarded. The algorithm was tested using data collected from a slickline tool with reference depth. Depth calculation using hydrostatic pressure is sensitive to measurement accuracy and density variations. Therefore, there is an error margin in the calculation. However, the erroneous depth measurement provides a window to evaluate magnetic field signal around expected collar depths that are known from the casing tally. High pass filtering and thresholding the windowed magnetic field, resulted in several potential peaks that could belong to a collar. Depending on the threshold level applied, several false positives or false negatives were found. Comparing the detected peak depth to the expected collar depth provided a Bayesian probability for the detection. Once a collar was detected with high probability, it was used as a reference depth for the subsequent collars. This limits the depth window size to evaluate magnetic fields. The algorithm identified 21 out of 26 collars within 1150 ft of depth in a steel cased test well filled with water. The calculated depth from the pressure was corrected at each detected collar. Compared to the reference depth from the slickline tool, the calculated depth stayed within 1 ft, even with 10% error margin in the depth calculation using hydrostatic pressure. The introduced method enables the miniaturization of the CCLs and paves the way for autonomous untethered downhole tools.

Introduction

Knowing the depth accurately is important for many downhole operations such as logging, sidetracking, or casing perforation. Even though the length of wire paid into the well provides a good primary reference for wireline and slickline tools, secondary methods such as casing collar location are still needed for improved accuracy. Upon completion of wells, a casing tally record is generated. This record shows the number of casing pipes along with the dimensions and sequence of each pipe used during the completion. Thus, it provides a good depth reference given that the pipe coupling, or collar, locations can be detected.

Casing collar locators (CCLs) are commonly integrated into downhole tools. Typically, CCLs are made of two strong permanent magnets and a coil or magnetometer (Seren and Deffenbaugh 2022). The magnets are placed at a distance from each other with the same type of poles pointing to each other. This creates a symmetry plane between the magnets where the normal component of the magnetic field becomes null due to the symmetry (Fig. 1a). The symmetry is broken when the magnetic permeability around one of the magnets changes. For example, a casing collar may have different inner and/or outer diameters than the steel pipes. This would break the magnetic symmetry if one of the magnets is in the steel pipe whereas the other one is within a collar making the normal component of the magnetic field on the symmetry plane nonzero (Fig. 1b). Thus, measuring the normal magnetic field component on the symmetry plane, one can detect casing collars.

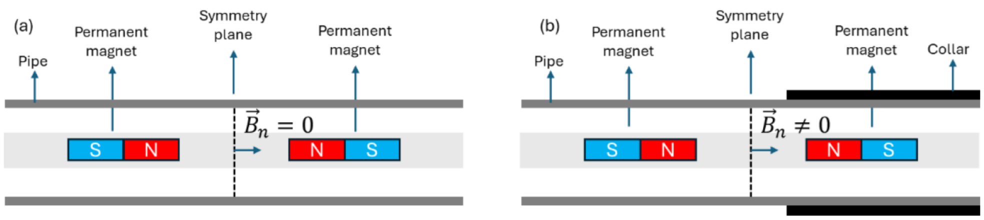

Figure 1—Casing collar locator working principle. (a) In a fully symmetrical case, the normal magnetic field component at the symmetry plane is zero; (b) When the symmetry is broken due to the pipe wall thickness, the normal magnetic field component at the symmetry plane becomes nonzero.

One area where CCLs can be of great benefit is untethered downhole logging tools (Seren et al. 2018, Sui et al. 2021, Hillman et al. 2023., Larbi Zeghlache et al. 2023). Having no wire to move, untethered tools can possibly reduce the need for heavy equipment such as cranes, winches, or blowout preventers (BOPs). Besides locomotion, the wire plays two other roles that are communication and depth control. Wireless communication in the downhole environment is challenging because of the attenuation of electromagnetic and acoustic signals. The untethered tools suggested in the literature are mostly autonomous for this reason. Untethered tools need alternative depth references such as CCLs (Santos 2015, Seren 2023). However, the requirement of strong magnets limits the use of CCLs to certain tool designs due to the weight of the magnets and the strong pull force generated towards the casing. It's desired to detect casing collars using a lightweight and low-power design that is suitable for miniaturized untethered logging tools.

One candidate to detect collars without using magnets is measuring the residual magnetic fields inside the steel casings. The steel pipes obtain a residual magnetization through their manufacturing and handling processes. The magnetic field usually shows a large amplitude swing around the ends of the pipes which can be detected using various types of magnetic sensors including magnetometer integrated circuits with form factors in mm's (Deffenbaugh and Seren 2025, Yoo 2002). However, the challenge then shifts to the signal to noise ratio. The magnetic signal acquired around the collars may resemble in amplitude and shape to arbitrarily changing magnetic fields within the steel pipes, which can be thought of as noise (Fig. 2). Here we approach this problem with the probabilistic methods and try to differentiate whether a detected magnetic signal is a collar or not.

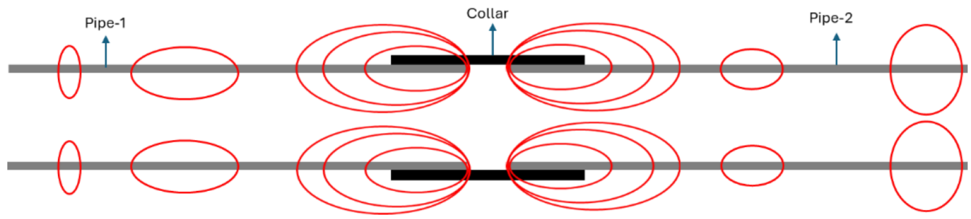

Figure 2—Cartoon illustration of residual magnetization along steel casing pipes and the collar between them

Probabilistic Casing Collar Locator Algorithm

Our algorithm is based on multi-hypothesis localization and Bayes filter (Thrun et al. 2002). The method starts with a mother hypothesis and then children hypotheses are generated as casing collars are detected. The hypotheses receive a score based on their likelihood and ones with the smallest likelihoods are pruned. The algorithm steps are summarized in Fig. 3.

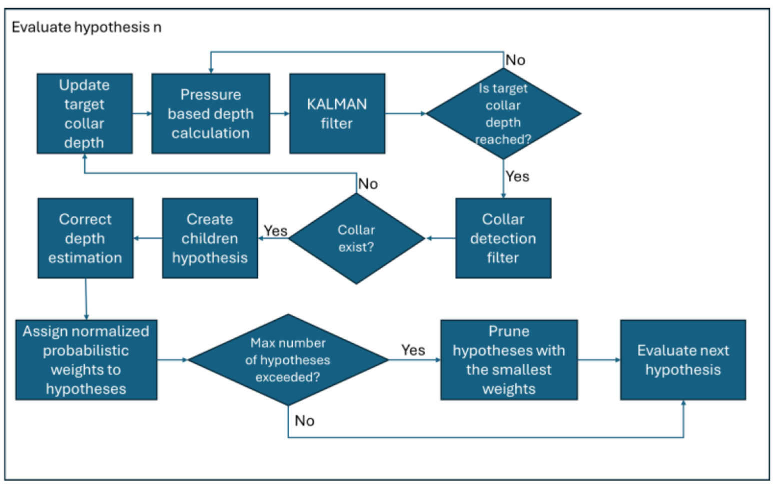

Figure 3—Probabilistic casing collar location algorithm

The method assumes that the depth can be estimated using hydrostatic pressure change and there exists a casing tally with the depths of collar. The mother hypothesis starts with a known depth (i.e., 0 ft), speed, estimated fluid density, and a likelihood weight of 1. As the tool travels, frequent pressure measurements are acquired. The pressure measurements are used together with the estimated fluid density to measure depth. This measurement has an error depending on the pressure transducer and the variance in fluid density and it tends to grow as the depth increases due to offsets in pressure and/or density. Optionally a Kalman filter is used if the pressure measurements are noisy. The Kalman filter can smooth the measurement noise by introducing a dynamic model. For example, if it's known that the tool usually travels at a constant speed, sudden changes in pressure can be omitted by the Kalman filter. Eventually a correction is required to the pressure based estimated depth to increase accuracy where casing collars play a critical role.

The algorithm frequently checks the estimated depth against the known collar depths from the casing tally. If the estimated depth passes a known collar depth the hypothesis believes that there should be collar in the last few feet. To test this belief, a measurement is accomplished by analyzing the axial magnetic field signal. The analysis finds fast changes in the signal by differentiating it with respect to depth and extracting absolute maximum (or maxima) above a threshold value. The maximum (or maxima) potentially indicates a casing collar. After this measurement, we end up having a believed location and a measured location for the collar. However, collar detections are also prone to error. Three example cases are shown in Fig. 4 where testing for collars in the vicinity of their tally depths produced a true location, a false location, and true and false locations simultaneously.

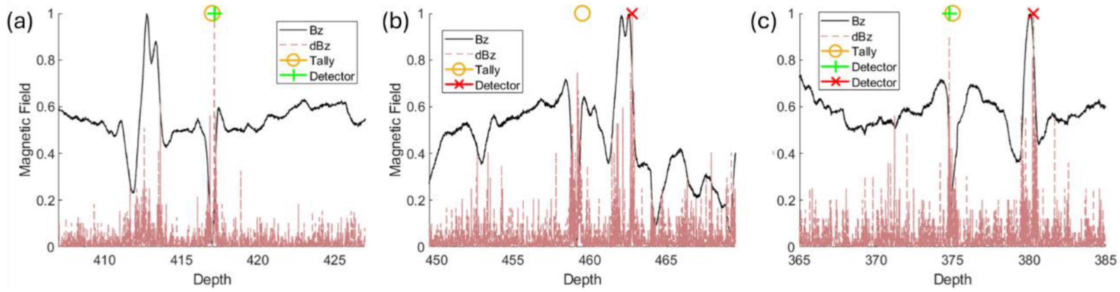

Figure 4—Examples of measured magnetic field distribution, change in magnetic field, and collar locations indicated by tally and our detector. (a) True detection; (b) False detection; (c) Mixed detection

For each collar detection, a child hypothesis is created that believes the detection is true. Its reference depth is assigned as the depth of the detected collar specified in the casing tally. For the new hypothesis, the historical depth assignments from the parent hypothesis' reference depth to the detected casing collar are reassigned by scaling these depths as shown in Eq. 1.

$$d^{m\prime} = d^t_{c,n-1} + \left(d^m - d^t_{c,n-1}\right)\frac{d^t_{c,n} - d^t_{c,n-1}}{d^m_{c,n} - d^t_{c,n-1}} \qquad \text{Eq. 1}$$

$$\alpha = \frac{d^t_{c,n} - d^t_{c,n-1}}{d^m_{c,n} - d^t_{c,n-1}} \qquad \text{Eq. 2}$$

$$\rho' = \rho / \alpha \qquad \text{Eq. 3}$$

where based on the correction made $d^{m\prime}$ is corrected measured depths, d^m is uncorrected measured depths, $d^m_{c,n}$ is the measured location of the nth collar, and $d^t_{c,n-1}$ and $d^t_{c,n}$ are the n-1st and nth collar depths specified by the casing tally. In our algorithm, we also correct the previously assumed density, ρ, by the coefficient α as shown in Eqs. 2 and 3.

As the tool travels through several casing collars, the number of hypotheses grow geometrically. This increases the computational cost. There needs to be a metric to tell which hypothesis is more reliable and prune the hypothesis with small probability. We address this by assigning a likelihood weight to the child and parent hypotheses after each detection event. The child inherits the likelihood weight from its parent scaled by a true rate (TR) factor as in Eq. 4. The parent continues to exist with the assumption that the collar detection was false, and its weight is scaled with a false rate (FR) as in Eq. 5.

$$w_{child} = \mathrm{TR} \times w_{parent,n} \qquad (4)$$

$$w_{parent,n+1} = \mathrm{FR} \times w_{parent,n} \qquad (5)$$

where n indicates the iterations happening at each detection event. With the assumption of there is a collar with a probability of 1 at the range of depths investigated, TR and FR are calculated using Bayesian probability that is a multiplication of initial belief of the casing collar that represented by a Gaussian probability distribution function, and true or false detection rates of the collar. This means, if the detected and believed locations are closer, the child hypothesis will receive a high weight and vice versa. Based on the tests in our test well, we found the true and false detection rates of the suggested collar detector as 0.9 and 0.1.

Experimental Validation

We integrated a magnetometer into a wireline tool and logged a steel cased test well. A CCL log using a commercial sensor has been previously run. The commercial CCL log was used to match the magnetic log. This served as the ground truth, or reference, for our experiments. Fig. 5 shows the two logs with respect to depth. It's clearly visible in this plot that at several collar peaks identified by the CCL log, there is also a corresponding sharp magnetic field change. The test well has a fiberglass section between 948 and 1034 ft and again after 1078 ft. The magnetic field peaks in the fiberglass sections belong to geophones behind the casing and not to casing collars. The conventional casing collar locator did not pick any signal around these peaks.

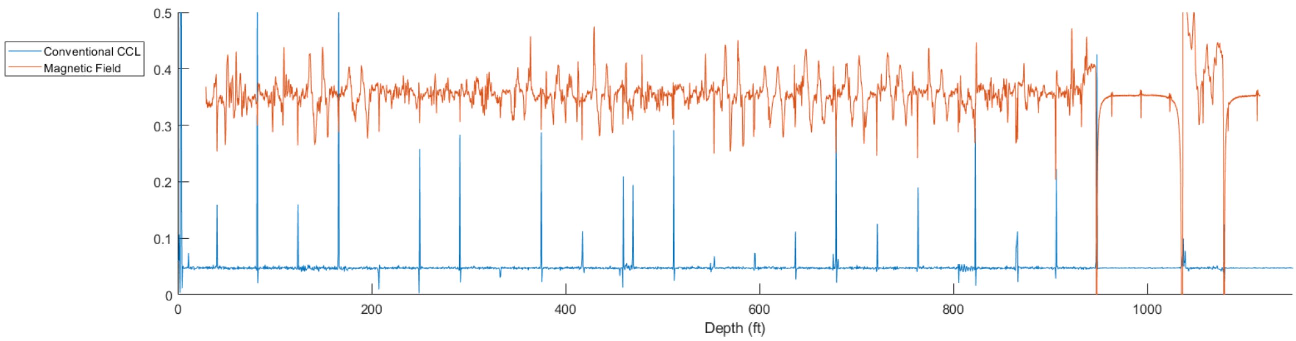

Figure 5—Conventional CCL log using a commercial sensor and the residual magnetic field of the test well

A pressure sensor was not available in our tool which is necessary for the algorithm to work. We calculated the pressure using hydrostatic pressure equation with model density of 1 g/cc. Gaussian noise with 0.1 psi standard deviation was added on the calculated pressure to mimic a realistic scenario. Fig. 6 shows an example section of the reference pressure calculated using the model density and the simulated pressure sensor output with the added noise.

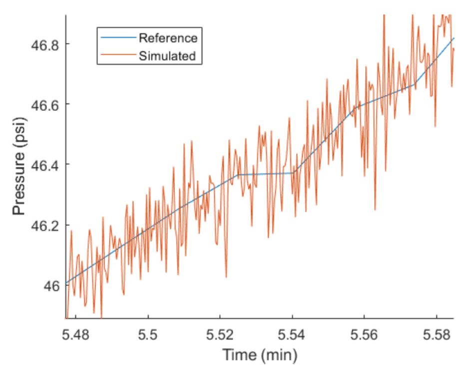

Figure 6—Example section of calculated reference pressure from wireline depth and simulated pressure sensor output with added noise on the reference pressure.

In practice, the density of the fluid may not be certainly known, and the algorithm should be able to handle this uncertainty. To test the algorithm's agility, the initial density of the initial hypothesis was set to 0.9 g/cc. +/- 0.3 g/cc change in the density was allowed. This would cause about 10% error in the depth calculation, putting the expected collar depth 4 ft away from its real location for the first collar, 8 ft for the second one and so on assuming 40-ft-long standard casing pipes.

Fig. 7a illustrates the results of the casing collar location algorithm with respect to the reference pressure and depth shown with the blue line where the pressure was derived from the wireline reference depth data using 1 g/cc liquid density. The black dashed lines are the boundaries set for the algorithm for stability. Effectively, it limits the possible fluid densities to an interval of 0.7 and 1.1 g/cc. The markers indicate casing collars detected by all the hypotheses generated during the algorithm. The algorithm starts from the shallowest measured depth and iteratively progresses towards the deeper locations while looking for collars. It can be seen that, initially, at shallower depths, collar detections are more spread. Thanks to the weightingbased evaluation of hypotheses, ones that are not consistently detecting the collars at the expected depths are eliminated and eventually moving towards deeper locations we are left with the hypotheses detecting collars aligned with the reference line. Fig. 7b shows the assumed density that was used by the hypothesis that received the highest score at the end. The density starts with the initial condition of 0.9 g/cc and finds its way to around 1 g/cc as collars are detected and depths, and assumed densities are scaled accordingly.

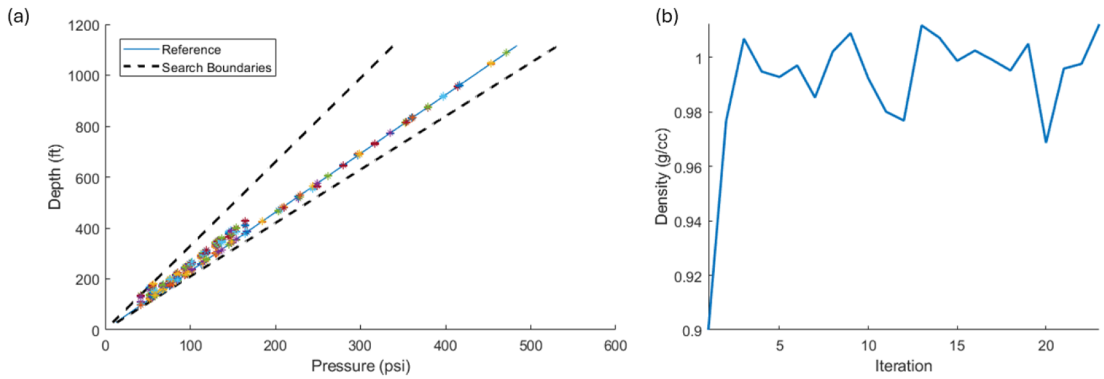

Figure 7—Experimental results of the probabilistic CCL algorithm: (a) Depth vs pressure plot showing the reference line, search boundaries, and detected collars by all hypotheses; (b) Assumed density by the best hypothesis to calculate depth from hydrostatic pressure over several iterations, or collar detections.

Fig. 8 shows the depth logs for pressure, magnetic field, and casing collars for reference and measurement by the best hypothesis. The reference collar depths are shown by orange circles. If a collar could not be detected, or false detection was assumed by the hypothesis, the circle is shown empty. If a true detection was assumed by the collar, a green plus was put in the circle if the detected collar was within 1-ft to its real location indicating possibly a true detection. If the detected collar was further than 1 ft to its real location, a red cross is placed indicating possibly a false detection. The best hypothesis guessed 21 out of 26 collars correctly, excluding signal at 0-ft that was used as the known reference location. 4 collars were missed and 1 collar was detected in false location, more than 1 ft away from its real location. In overall, the measurement depths agree very well with the reference in both pressure and magnetic field plots. Fig. 9 shows the error in depth measurement. It mostly stayed within +/- 1 ft except around the false collar detection at about 600 ft.

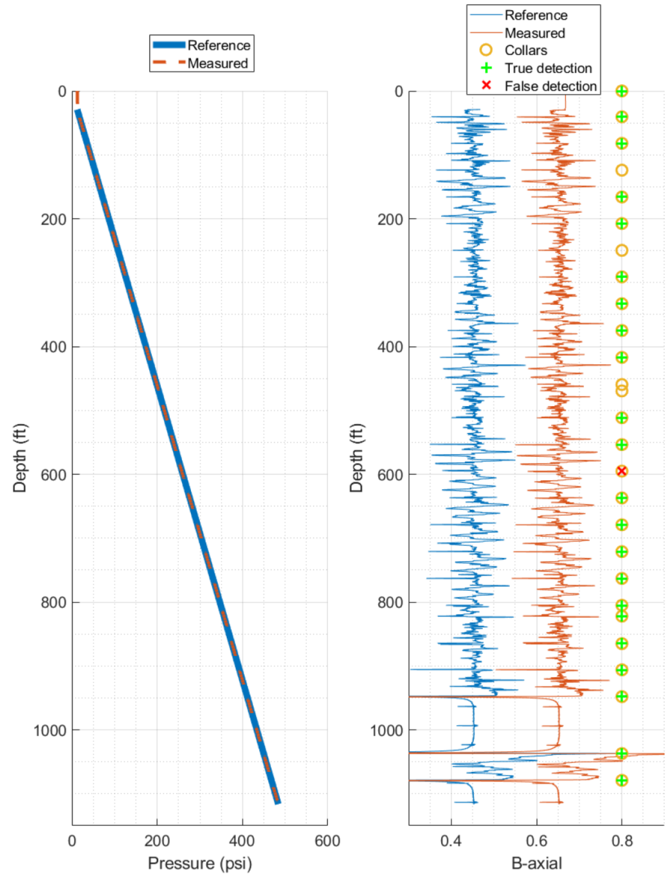

Figure 8—Depth logs for pressure, magnetic field and casing collars showing both reference and algorithm based measurement data.

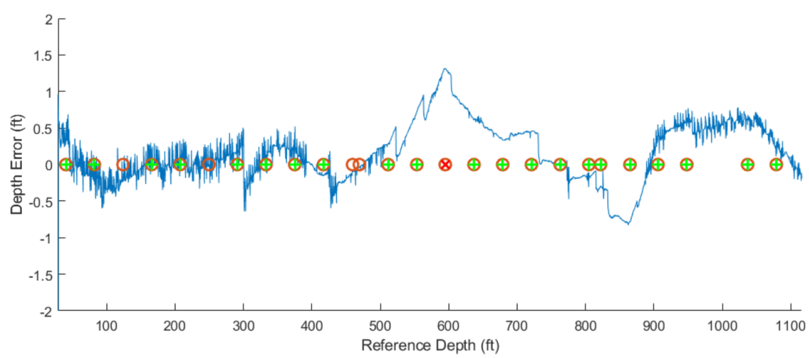

Figure 9—Depth measurement error and casing collars. Empty circles, green plus signs, and red cross signs indicate missed, true, and false detections, respectively.

Conclusions

The CCL is a staple downhole tool widely used for depth detection in steel-cased hole logging operations. Traditional CCLs rely on strong magnets, which can limit their applicability—particularly in untethered downhole tools. However, advancements in microelectronics and sensor technologies are opening new avenues for enhancing these tools.

In our approach, we utilized a magnetometer integrated circuit to detect residual magnetic fields induced in steel casing pipes. We developed an algorithm to identify casing collars based on the rapid changes in magnetic field strength around them. The presence of numerous peaks and valleys in the magnetic field within steel pipes makes it difficult to reliably detect collars using magnetic signatures alone, reducing detection certainty.

To address this, our algorithm incorporates additional data from a pressure sensor, filtered through a Kalman filter, to improve collar detection accuracy. We also employed the multiple hypotheses theorem to further enhance reliability.

Our experiments, conducted using a slickline tool in a test well, yielded promising results: 21 out of 26 collars were correctly identified. The calculated depth error remained largely within ±1 foot—an important threshold for making critical decisions based on depth log data.

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